

MATH 180 Graph Theory — Discussion 8

Counting labeled trees by degree sequence (Ch. 8.2)

Quick Recall (Ch. 8.1–8.2)

Spanning trees & notation. For a graph G , let $T(G)$ denote the number of spanning trees of G . Throughout, we assume the vertex set is $[n] = \{1, 2, \dots, n\}$.

Cayley's formula (Thm 8.1.1). For each $n \geq 2$, the number $T(K_n)$, i.e. the number of trees on given n vertices, equals n^{n-2} .

Handshake lemma in trees. Any tree on n vertices has exactly $n - 1$ edges, hence

$$\sum_{i=1}^n d_i = 2(n-1),$$

where d_i is the degree of vertex i .

Multinomial coefficient. The number of ways to partition $[n]$ into ordered groups of sizes $k_1, \dots, k_r$ (with $k_1 + \dots + k_r = n$) is

$$\binom{n}{k_1, k_2, \dots, k_r} = \frac{n!}{k_1! k_2! \dots k_r!}.$$

Proposition 8.2.1 (score-counting). Let $d_1, d_2, \dots, d_n$ be positive integers summing up to $2n - 2$. Then the number of spanning trees of the graph K_n in which the vertex i has degree exactly d_i for all $i = 1, 2, \dots, n$ equals

$$\frac{(n-2)!}{(d_1-1)! (d_2-1)! \dots (d_n-1)!}.$$

Proof idea. Induction on n : pick a leaf, remove it, and use a bijection onto trees of K_{n-1} with the decremented degree sequence.

Mini-example ($n = 4$, score $(3, 1, 1, 1)$). By the proposition,

$$\frac{2!}{2! 0! 0! 0!} = 1$$

spanning tree of K_4 has vertex 1 of degree 3 – the star centered at 1.

Practice: Textbook Exercise 8.2.1* — trees with degrees in $\{1, 3\}$, part (a)

(a)* Find the number of trees (on a given vertex set $[n]$) in which all vertices have degree 1 or 3.

Hints

Suggested steps. (1) Let k be the number of degree-3 vertices, so $n - k$ vertices are leaves. Use the handshake lemma $\sum d_i = 2(n - 1)$ to solve for k in terms of n . (2) Observe what parity of n is forced. (3) Choose which vertices are degree-3 in $\binom{n}{k}$ ways; for each such choice, the degree sequence is fixed, so apply Proposition 8.2.1.

Answer: Step 1: Use handshake to determine k . Let k denote the number of degree-3 vertices, and let $n - k$ be the number of degree-1 vertices (leaves). By the handshake lemma,

$$\sum_{i=1}^n d_i = 3k + 1 \cdot (n - k) = 2k + n = 2(n - 1) = 2n - 2,$$

which gives $k = \frac{n-2}{2}$.

Step 2: Parity restriction. Since $k = (n - 2)/2$ must be a nonnegative integer, n must be *even* (and $n \geq 2$). For n odd, no such tree exists, so the count is 0.

Step 3: Count for n even. Fix n even and set $k = (n - 2)/2$. There are two layers of choice:

- Choose which k vertices have degree 3: $\binom{n}{k}$ ways. This pins down the score $(d_1, \dots, d_n)$.
- Count trees with that score: by Proposition 8.2.1,

$$\frac{(n-2)!}{(3-1)!^k \cdot (1-1)!^{n-k}} = \frac{(n-2)!}{2^k}$$

(since $(3 - 1)! = 2$ and $(1 - 1)! = 0! = 1$).

Final answer.

$$\#\{\text{such trees}\} = \begin{cases} \binom{n}{(n-2)/2} \cdot \frac{(n-2)!}{2^{(n-2)/2}} & \text{if } n \text{ even,} \\ 0 & \text{if } n \text{ odd.} \end{cases}$$

□

Sanity check ($n = 4$). Formula gives $\binom{4}{1} \cdot \frac{2!}{2^1} = 4 \cdot 1 = 4$, which matches the four labeled stars $K_{1,3}$ on 4 vertices (one for each choice of center).

(b)* What if we allow degrees 1, 2, or 3?

💡 Hints

Suggested steps. (1) Let k be the number of degree-3 vertices and ℓ the number of degree-2 vertices; the remaining $n - k - \ell$ vertices are leaves. Use the handshake lemma to express ℓ in terms of k and n . (2) Determine which values of k are admissible (need $\ell \geq 0$ and $k \geq 0$). (3) For each admissible k : count the number of ways to choose which vertices play each role (multinomial coefficient), then apply Proposition 8.2.1. Sum over k .

Answer: Step 1: Parametrize by k . Let k = number of degree-3 vertices, ℓ = number of degree-2 vertices, so $n - k - \ell$ vertices are leaves. By the handshake lemma,

$$3k + 2\ell + 1 \cdot (n - k - \ell) = 2k + \ell + n = 2n - 2,$$

which gives $\ell = n - 2 - 2k$. Hence the number of leaves is $n - k - \ell = k + 2$.

Step 2: Admissible range. We need $k \geq 0$ and $\ell = n - 2 - 2k \geq 0$, i.e. $0 \leq k \leq \lfloor \frac{n-2}{2} \rfloor$. Note (b) allows *both* parities of n (compare with (a), which forced n even).

Step 3: Count for fixed k . Two layers of choice, exactly as in part (a):

- *Assign vertices to roles:* choose which k vertices have degree 3, which $\ell = n - 2 - 2k$ have degree 2, and which $k + 2$ are leaves. This is the multinomial

$$\binom{n}{k, n - 2 - 2k, k + 2} = \frac{n!}{k! (n - 2 - 2k)! (k + 2)!}.$$

- *Count trees with that score:* by Proposition 8.2.1,

$$\frac{(n - 2)!}{(3 - 1)!^k (2 - 1)!^\ell (1 - 1)!^{k+2}} = \frac{(n - 2)!}{2^k}$$

(degree-2 and degree-1 vertices each contribute factors of 1).

Step 4: Sum over k .

$$\#\{\text{such trees}\} = \sum_{k=0}^{\lfloor (n-2)/2 \rfloor} \frac{n! \cdot (n - 2)!}{k! (n - 2 - 2k)! (k + 2)! \cdot 2^k}.$$

□

Sanity check ($n = 4$).

- $k = 0$: $\ell = 2$, leaves = 2. Contribution = $\frac{4! \cdot 2!}{0! \cdot 2! \cdot 2! \cdot 1} = 12$. (These are the 12 labeled paths P_4 .)
- $k = 1$: $\ell = 0$, leaves = 3. Contribution = $\frac{4! \cdot 2!}{1! \cdot 0! \cdot 3! \cdot 2} = 4$. (The 4 labeled stars $K_{1,3}$.)

Total: $12 + 4 = 16 = 4^{4-2} = T(K_4)$. ✓ This is no accident: every tree on 4 vertices has maximum degree ≤ 3 , so allowing degrees in $\{1, 2, 3\}$ recovers *all* spanning trees, and the sum reduces (a special case of) the score-summation derivation of Cayley's formula given in lecture.

Remark. A similar sanity check for $n = 5$ gives $60 + 60 = 120 = 5^3 - 5 = T(K_5) - \#\{\text{stars } K_{1,4}\}$, since the only tree on 5 vertices with a vertex of degree ≥ 4 is the star.